Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/07/2004
PI Name:	Davis, Robert Ph.D.		
Project Title:	Surface Collisions Involving Particles and Moisture (SCIP'M)		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80309-0424	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	06/01/2000	**End Date:**	05/31/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC3-796		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Ground-based experiments: The initial experiments will be performed at the University of Colorado, both to test the experimental design and to obtain data on high-speed collisions. Flight experiments: The subsequent experiments will be performed in a low gravity environment, such as provided by parabolic flight, to obtain data on collisions with small impact speeds and angles. Theoretical program: Supporting theory will be developed to interpret the experimental findings and determine the relative contributions of liquid lubrication forces and solid elastic and friction forces during impact. Determine the conditions for when a colliding particle sticks to a wet surface instead of rebounding. For rebound, determine how the receding angle and velocities relate to the approaching angle and velocities. Determine what forces and torques occur during a collision, and determine to what extent they can be described by the classical lubrication, elastic, and friction theories. Determine when splashing occurs during a collision. The experimental program will include both ground-based and KC-135 gravitational levels. Theoretical analysis will be used to interpret		

	the experimental findings. The task provides fundamental understanding of particle collisions with surfaces (or particles) when both solid mechanics and fluid mechanics play important roles. It also is relevant to predicting suspension flows, and it can be applied to wet granular materials flows. The knowledge will also be used to help understand agglomeration processes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	1. Experiments with Head-on Collision: Initial ground-based laboratory experiments were performed to determine the conditions under which a small particle would stick as opposed to bounce when dropped on a wetted surface. Using a camera and strobe light, these experiments were extended to determine the coefficient of restitution (rebound velocity divided by impact velocity) in those cases where rebound occurred. It was found that a critical impact velocity must be exceeded for bouncing to occur and that the coefficient of restitution increases with increasing thickness and viscosity of the fluid layer. The data have been correlated in terms of the Stokes number, which is a dimensionless ratio of particle inertia to viscous forces. The critical Stokes number for rebound and the coefficient of restitution also depend weakly on an elasticity parameter, which is a dimensionless ratio of viscous deforming forces to elastic restoring forces. A simple theory for predicting the critical Stokes number and the coefficient of restitution has also been developed, based on lubrication theory for the viscous losses in the narrow gap and Hertzian contact theory for the elastic deformation and rebound. It is in good agreement with experiments. The results were presented as a poster in July 2001 at the Gordon Research Conference on Gravitational Effects in Physicochemical Systems. A journal paper (Davis, Rager, and Good, 2002, JFM 468:107-119) has been published on this work. More recent experiments have examined head-on collisions of a ball with a hard surface or a soft surface, both dry and wet. The soft surface is composed of a thin cloth or fabric on top of a quartz disk. In addition to using low gravity (see item 4), low-speed collisions are obtained using a pendulum. In contrast to a hard surface, the soft surface gives rise to a reduction in the coefficient of restitution with increasing impact velocity. This finding is due to the inelastic losses in the deformable fabric and the viscous losses from the squeeze flow on the porous fabric during the collision. 2. Experiments with Collisions at an Angle: Additional experiments were performed where the impact velocity was not normal to the plane of the wetted surface. Perhaps surprisingly, increasing the angle of inclination of the surface from horizontal increased the overall coefficient of restitution at the same impact velocity or Stokes number. The tangential coefficient of restitution was found to be close to unity (i.e., only a small reduction in the tangential velocity due to collisions was observed), while the normal coefficient of restitution is considerably less than unity due to viscous losses and becomes smaller as the normal impact velocity becomes smaller. As a result, the angle of rebound with respect to the surface is less than the angle of impact. A manuscript on this work has been published (Kantak & Davis, 2004, JFM 509: 63-81). 3. Model of Collisions at an Angle: When a particle collides with a wet plane at an angle other than 90 degrees (i.e., at an angle besides normal to the surface), tangential forces will occur. These tangential forces are expected to include both lubrication forces and potentially solid-solid contact friction if either or both surfaces have microscopic surface roughness. To begin to understand these forces, experiments were performed and a model was developed for a rough sphere rotating and translating down a smooth, inclined plane (i.e., not colliding or bouncing). This work has helped the development of a fundamental model of the interaction, including both normal and tangential forces (Galvin, Zhao, and Davis, 2001 Phys. Fluids 13: 3108-3120). Continued work has been performed with an artificially roughened plane, showing good agreement between theory and experiment for the translational and rotational velocities of the sphere, as well as its effective hydrodynamic separation, as it moves down the plane and encounters the roughness elements (Zhao, Galvin, and Davis, 2002, IJMF 28: 1787-1800; Davis, Zhao, Galvin, and Wilson, 2003, Phil. Trans. 361: 871-894). More recent modeling of oblique collisions of a sphere with a surface has considered the sliding lubrication friction impacted on the ball by the thin fluid layer on the surface. The resulting force and torque give rise to a slight reduction in the tangential velocity of the sphere and cause rotation (Kantak and Davis, 2004, JFM 509: 63-81). 4. Parabolic Flight Experiments: The first set of flight experiments on a KC-135 aircraft were performed in July 2003 out of Cleveland, OH. Experiments were performed at low impact velocities (2-24 cm/s) that are not possible in normal gravity. Thinner and less viscous fluid layers were used in these flights than on earth. Rebound was observed as expected at the higher velocities, but only limited data were obtained at the lower velocities due to small accelerations in the aircraft causing the balls to sometimes miss the target. A second round of KC-135 experiments with shorter distances from launch to target was performed out of Houston, TX, in October 2003. These experiments gave additional data for low-speed collisions of a sphere with a hard surface overlaid with a thin fluid layer of low viscosity. Additional experiments were performed for low-speed collisions of a sphere with a cloth-covered surface. Without the cloth, the wet coefficient of restitution increased with increasing impact velocity, whereas the opposite was found to occur with the cloth.
Bibliography Type:	Description: (Last Updated: 12/12/2005)
Articles in Peer-reviewed Journals	Davis, R. H.; Zhao, Y.; Galvin, K.; Wilson, H. "Solid-Solid Contacts Due to Surface Roughness and Their Effects on Suspension Behavior" Phil. Trans. R. Soc. Lond. 361, 871-894 , Mar-2003
Articles in Peer-reviewed Journals	Kantak, A. A.; Davis, R. H. "Oblique Collisions and Rebound of Spheres from a Wetted Surface" J. Fluid Mech. 509, 63-81 , Jun-2004